

Bishop's University
Department of Computer Science
Intelligent systems and Neural Nets – CS504
FL2024 – Project

Problem: Breast Cancer Detection using Ensemble Classifiers in Python/R.

Dataset:

- Breast Cancer Wisconsin (Diagnostic) Data Set
- <https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data/data>

Classification Model Technique: Combine Logistic Regression (LR), SVM (Support Vector Machine), PCA and NFM for breast cancer detection.

Tasks:

Divide the dataset into three subsets: **Training (60%)**, **Validation (20%)**, and **Test (20%)**. Use the Training and Validation sets to train the global model (as described in Figure 1) and reserve the Test set for evaluating the model's performance.

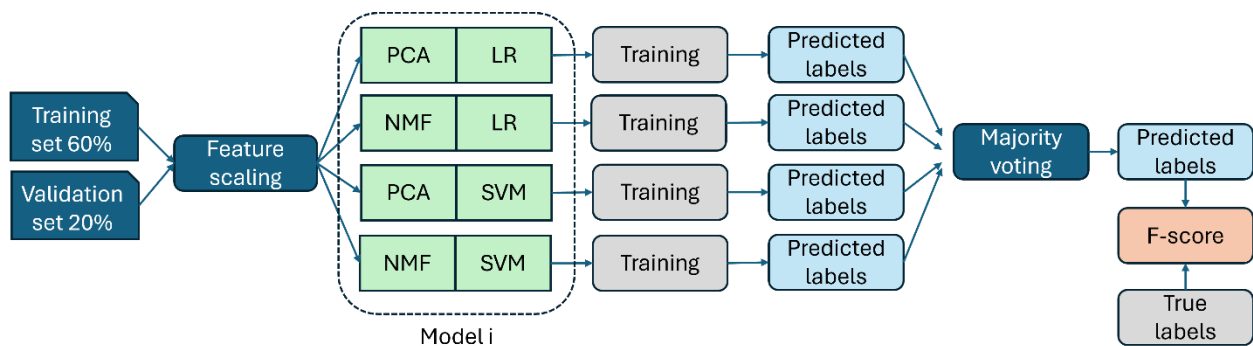


Figure 1: Training process

I. Given the training and validation set:

1. **Apply Feature Scaling:** Use Min-Max scaling on both the training and validation sets.
2. **Feature Extraction:** Extract features from the data using PCA and NMF, ensuring the same number of components is used for both methods.
3. **Train Classifiers:** Train each classifier individually using cross-validation on the training set, applying default parameters for each model.
4. **Predict Labels:** Apply the trained classifiers to the validation set to obtain predicted labels.
5. **Majority Voting:** Implement majority voting to determine unified predicted labels.
6. **Compute F-Score:** Calculate the F-score using the predicted labels compared to the true labels of the validation set.
7. **Iterate with Different Components:** Repeat steps 2 to 6 for varying numbers of components (2, 4, 6, 8, and 10).
8. **Report Best F-Score:** Identify and report the optimal number of components based on the highest F-score.

9. **Model Configuration:** You will develop five models: model_2, model_4, model_6, model_8, and model_10, where each model corresponds to a specific number of feature components used during training. For instance, model_4 is trained with four feature components. This model is an ensemble of five different combinations of feature extraction and classifier methods, represented as follows:
 $\text{model_4} = \{(\text{PCA_4}, \text{LR_4}), (\text{NMF_4}, \text{LR_4}), (\text{PCA_4}, \text{SVM_4}), (\text{NMF_4}, \text{SVM_4})\}.$
10. **Apply the Best Model:** Use the best global model (Model_best) on the test set to compute the corresponding F-score. The testing model is outlined below.
 - Ensure that the test set is feature-scaled using the same scaler applied to the training set.

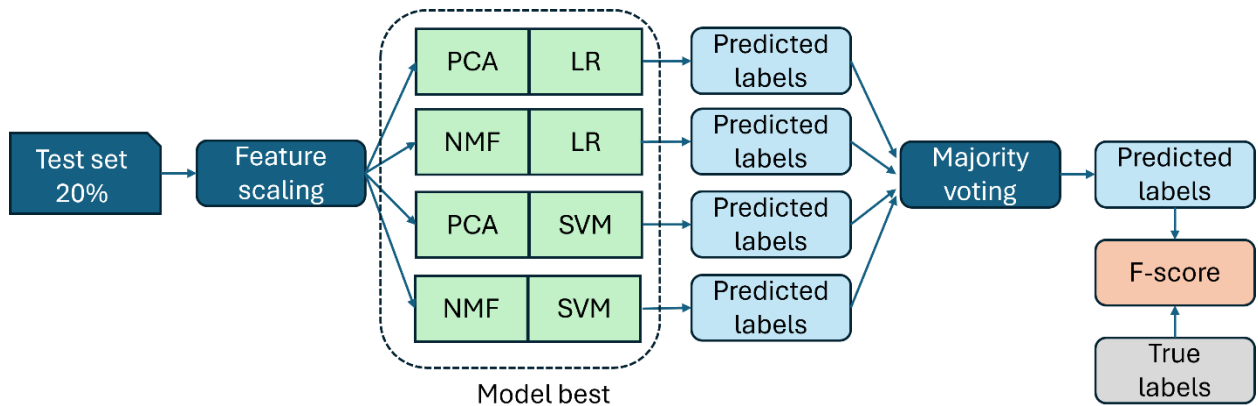


Figure 2: Test process

II. Report Structure. provide a report according to the following structure:

1. Title of the Project
2. Authors and Affiliations
3. Abstract
4. Introduction
5. Proposed Method
6. Dataset Description
7. Data Exploration:
 - Present statistics on data distribution.
 - Analyze feature correlations and discuss the implications.
8. Results: Report F-scores for each model (model_2, model_4, ..., model_10) using tables and bar plots. Discuss the findings.
9. Test Set Data Distribution:

Visualize the data distribution of the test set using the first two components of PCA and NMF and discuss the findings.

Delivery:

1. **Submit a Jupyter Notebook** that is clearly organized and well-structured.
2. **Provide a report** of approximately five pages, ensuring it is clear and well-organized. The organization and clarity of the report will be graded.